Molecular interactions in the context of Alzheimers disease

Annotation rich dataset from the IMEx Consortium

EMBL-EBI

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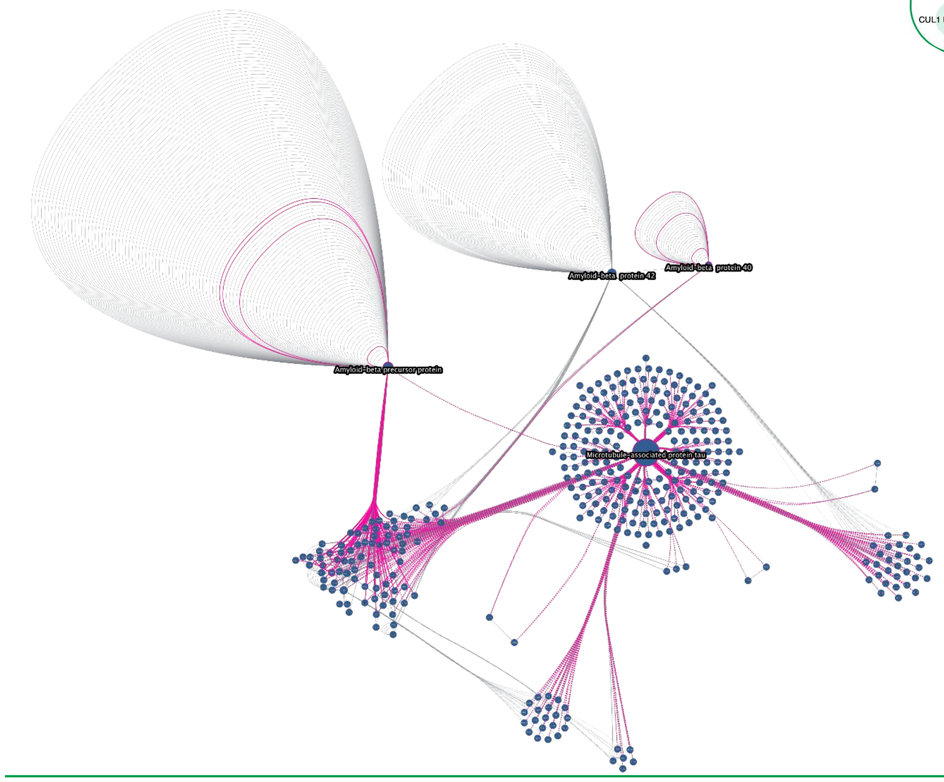
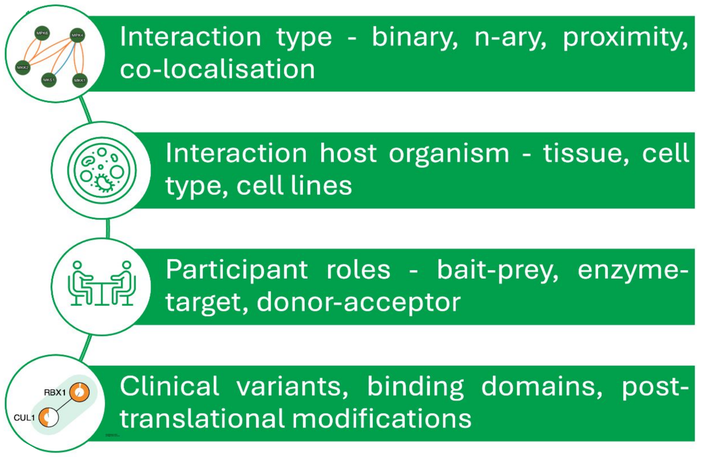
International Molecular Exchange Consortium[1]

https://www.imexconsortium.org/

- **Consortium of of open** resources of **manually curated** molecular interactions from **scientific literature**
- Curation strategy - **deep curation model** aimed to reflect **most accurate** representation of **experimental evidence**
- Users can explore the **interaction dynamics** in diverse contexts like **clinical variant**-specificity, **tissue** specificity, response to **external conditions**
  - **IntAct**, one of the consortium partners provides **advanced data filter options** to access the multi layered **interaction data** and **network visualisation**





IMEx's Alzheimers dataset

- Encompassing **63,000** interactions - binary interactions detected by 2-hybrid, enzymatic assays; **spoke-expanded binaries** detected by AP-MS
- **51,000** interactions have MI Score 0.5 or higher - reflecting the confidence of interactions captured.
- **6,000 interactions** were captured which were **detected** in brain-associated tissues such as **neurons, hippocampus, and cortex**
- More than **6,000** interactions affected by **clinical variant/mutation feature**

IntAct[2] - Data access enabled by advanced filter options

https://www.ebi.ac.uk/intact

IntAct's advanced filter options allows users to **explore data** at multiple layers including the **host organism**, the interaction type, the detection method, interactor **species**

Network of human protein-protein interactions detected in human brain-related tissues in the context of Alzheimers

Network generated by IntAct Cytocape app[3] using the advanced filter options

- **Nodes and edges** affected by **clinical variant/mutation** highlighted in pink
- Interactions with **confidence** score of **0.5 and above** were chosen
- **Loops** represent **oligomerization**/self-interactions

www.ebi.ac.uk

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3. Ragueneau E, Shrivastava A, Morris JH, et al. **IntAct App: a Cytoscape application for molecular interaction network visualization and analysis.** Bioinformatics (Oxford, England). 2021 Oct;37(20):3684-3685. DOI: 10.1093/bioinformatics/btab319. PMID: 33961020; PMCID: PMC8545338.

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